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Design of object-oriented databases using UML

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LIS – 3012

Unified Modeling Language (UML)

- Graphical language used for specify, build, and document software artifacts
- Result of a unifying action over distinct object-oriented modeling languages:
  - Allows to model systems from the conception phase to the execution phase
  - Can be interpreted by humans and machines
- First release - 1997

UML conceptual model

- Vocabulary that includes templates/designs for:
  - Elements: they permit to modelate software artifacts
    - Structure
    - Behavior
  - Relationships: they associate elements among them
  - Diagrams: they group collections of elements

- A well-formed UML model consider rules for:
  - Constructor names
  - Context that provides meaning to a name
  - Visibility for specifying how to use names
  - Integrity for specifying how to relate elements consistently
  - Execution for specifying what it means to run/simulate a model

UML diagram

- System:
  - Collection of subsystems organized for a specific purpose
  - A subsystem is a bundle of elements where some specify the behavior of others
  - It is described by several models from different perspectives (dynamic and static)
- Model:
  - Represent a complete and descriptive simplification of reality
  - Allow a better understanding of a system.
- Diagram:
  - Graphical representation of a set of documents that conforms a perspective

UML diagram types

- Static perspective:
  - Classes: show a set of classes, interfaces and relationships
  - Objects:
    - Show a set of objects and relationships
    - Display the database status on a time t
  - Components:
    - Show the set of component s and relationships
    - Display the static point of view of a system implementation
  - Installation:
    - Show a set of nodes and relationships
    - Display the static perspective of a system architecture installation

- Dynamic perspective:
  - User case: Organize the system behaviors
  - Sequence: temporal order of messages
  - Collaboration: structural organization of objects that send and receive messages
  - State: system state changes given events production
  - Activities: flow-control between activities

Class diagrams

- Classes, relations, and dependencies
- Model the static perspective of a system design
- Class definition:
  - Name
  - Attributes
  - Methods

<Class name>
<attribute$_1$>: Type ... <attribute$_n$>: Type
<method$_1$>(<Parameter list>) ... <method$_n$>(<Parameter list>)

Classes features

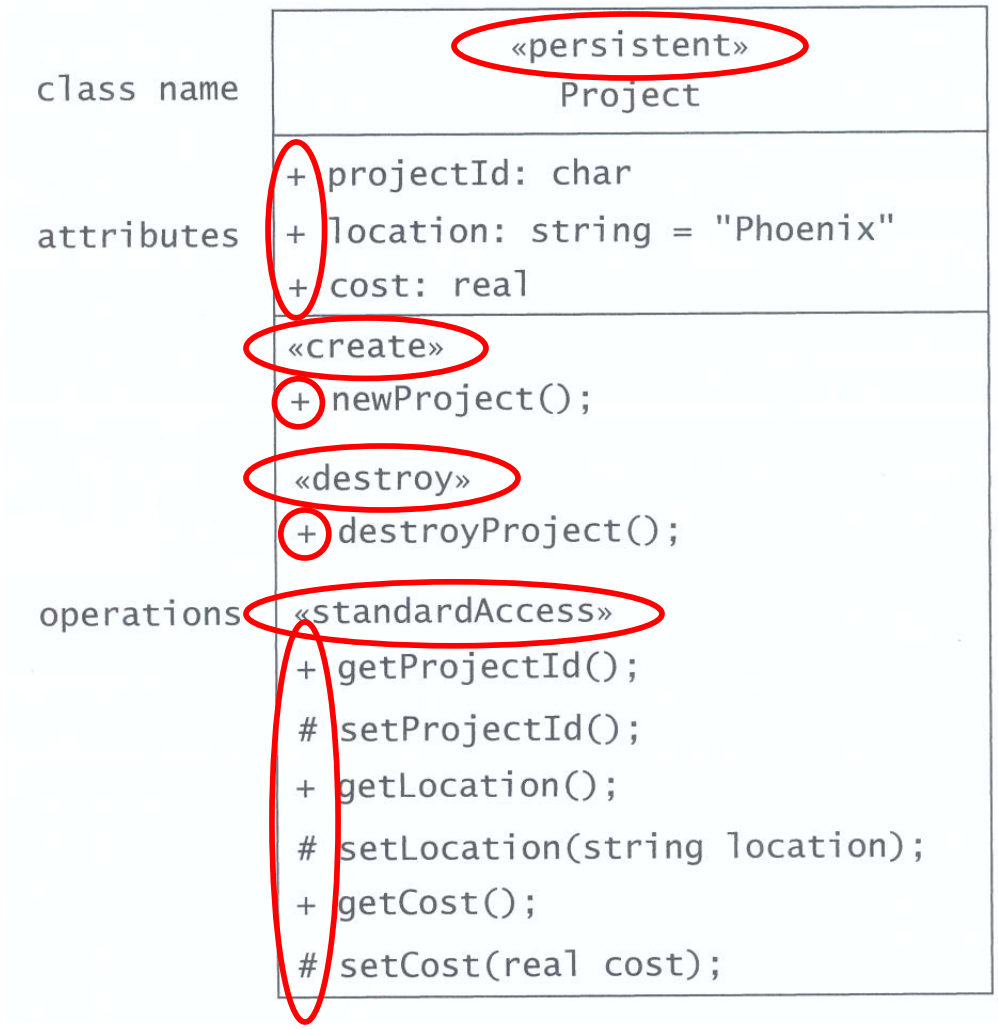
Attributes and methods

[<< Stereotype >>] [Visibility] [/] <Attribute name>
[Multiplicity] : [type] [= initial value]

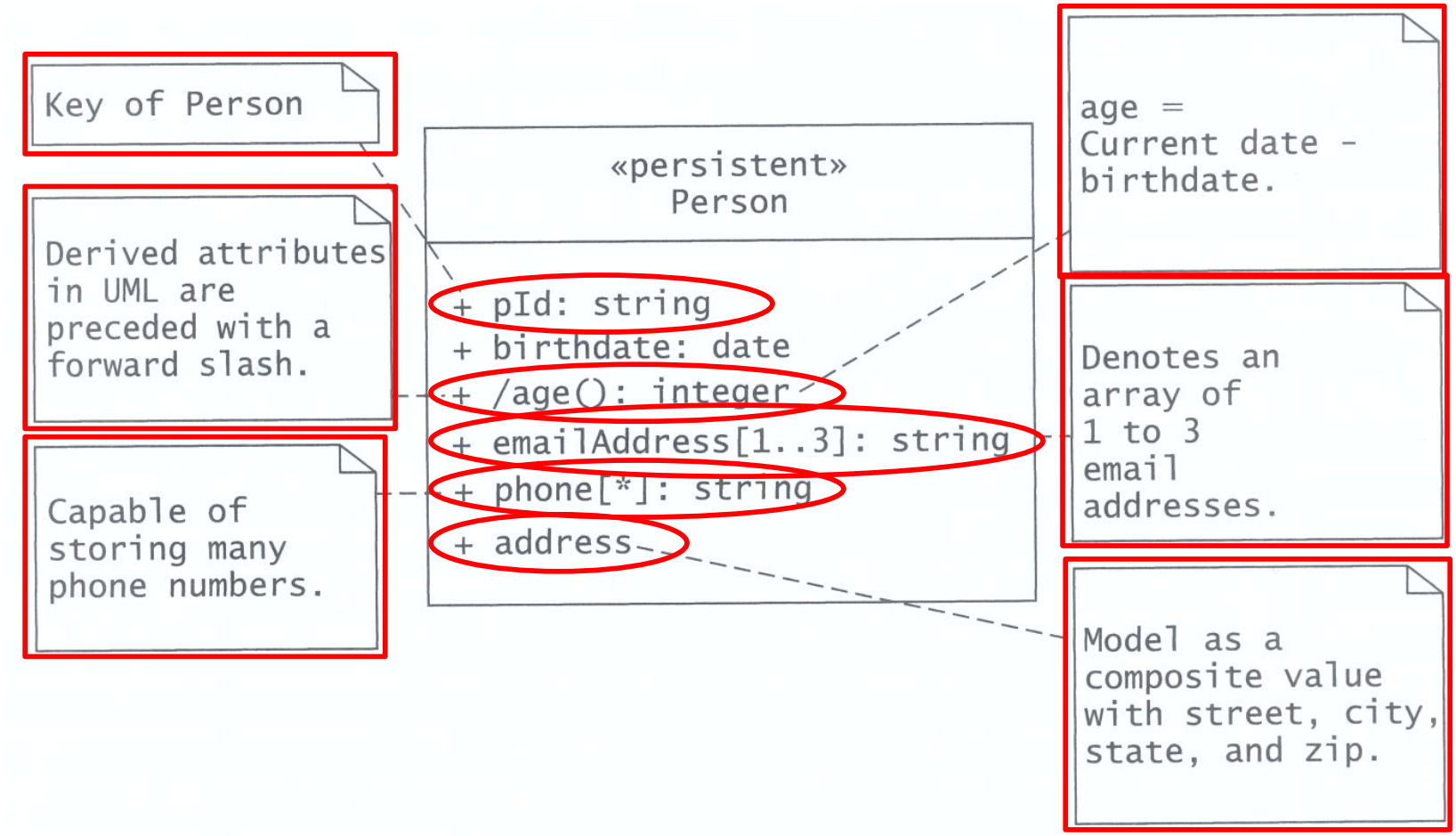
[<< Stereotype >>] [Visibility] <Method name> (
[Parameters list]) : [Result type]

Visibility	Use
+	Attribute or public method
#	Attribute or protected method
–	Attribute or private method

Stereotypes: Examples



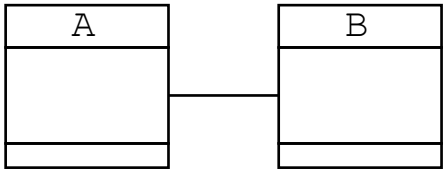
UML notes/comments



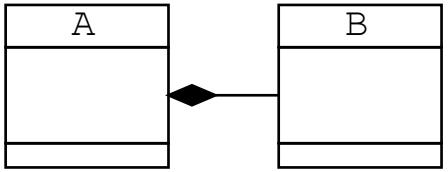
Classes and relationships

Classes and relationships

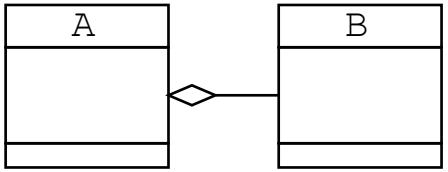
- Types of bi-directional relations (1:1, 1:N, N:M):



- Association:
  - Class A is associated to class B (Employee --- Branch)
  - Both classes exist



- Composition
  - Class A contains class B (Car ◆--- Tire)
  - A is the whole entity and B is just a part



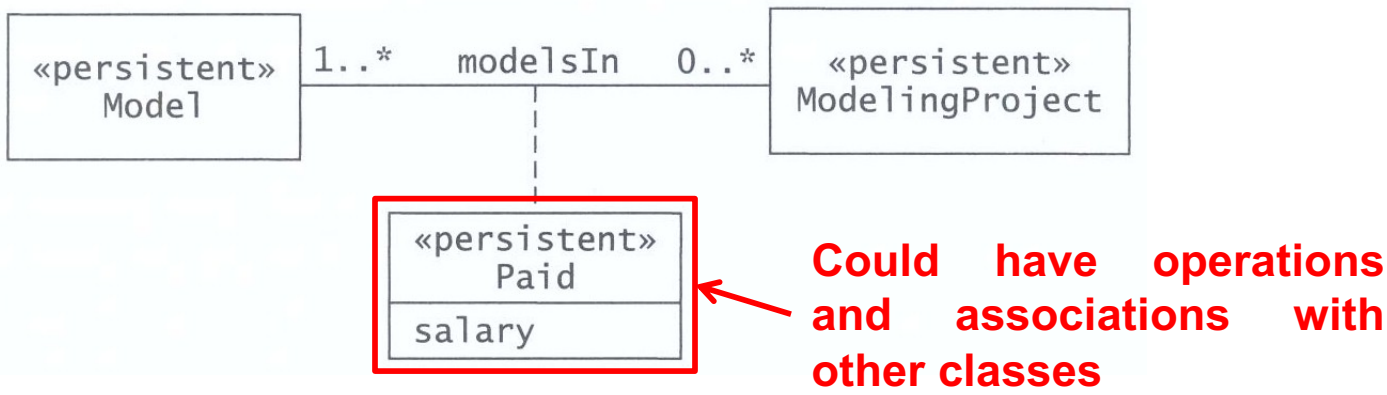
- Aggregation:
  - Class A it is compose of class B
  - (Company ◇--- Employee)

Associations: Terminology

Name	sponsors	The arrow indicates the way to read the association
Roles	sponsoredBy	A modeling project is supported by a sponsor
	projectsSponsored	A sponsor has many projects
Cardinality	1..1	A modeling project must be supported by one and only one sponsor
	0..*	A sponsor can support zero or many projects

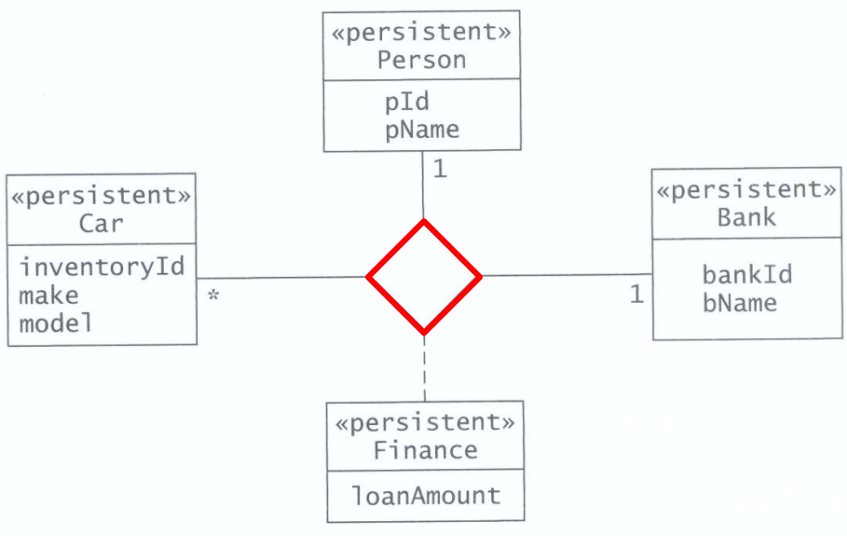
Association classes

- Ejemplo:

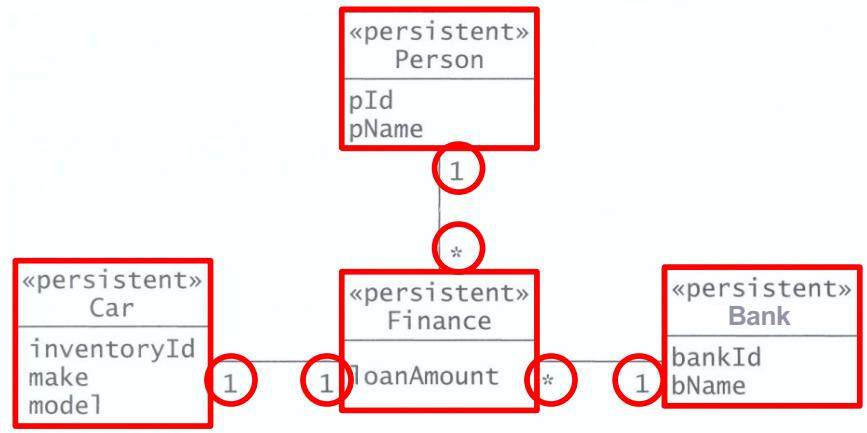


- Asociación materializada (*reification*):

N-ary associations



Ternary association



Materialized association